The greater the amplitude and the shorter the wavelength, the greater the energy. This reflects what you would expect intuitively—more frantic vibrational patterns have more energy, while less frantic ones have less energy. We give a couple of examples in Figure 6.3. This is again familiar, as violin strings that are plucked more vigorously will vibrate more wildly, while those plucked more gingerly will vibrate more gently. Now, from special relativity we know that energy and mass are two sides of the same coin: Greater energy means greater mass, and vice versa. Thus, according to string theory, the mass of an elementary particle is determined by the energy of the vibrational pattern of its internal string. Heavier particles have internal strings that vibrate more energetically, while lighter particles have internal strings that vibrate less energetically.

Since the mass of a particle determines its gravitational properties, we see that there is a direct association between the pattern of string vibration and a particle's response to the gravitational force. Although the reasoning involved is somewhat more abstract, physicists have found that a similar alignment exists between other detailed aspects of a string's pattern of vibration and its properties vis à vis other forces. The electric charge, the weak charge, and the strong charge carried by a particular string, for instance, are determined by the precise way it vibrates. Moreover, exactly the same idea holds for the messenger particles themselves. Particles like photons, weak gauge bosons, and gluons are yet other resonant patterns of string vibration. And of particular importance, among the vibrational string patterns, one matches perfectly the properties of the graviton, ensuring that gravity is an integral part of string theory.7

So we see that, according to string theory, the observed properties of each elementary particle arise because its internal string undergoes a particular resonant vibrational pattern. This perspective differs sharply from that espoused by physicists before the discovery of string theory; in the earlier perspective the differences among the fundamental particles were explained by saying that, in effect, each particle species was "cut from a different fabric." Although each particle was viewed as elementary, the kind of "stuff" each embodied was thought to be different. Electron "stuff," for example, had negative electric charge, while neutrino "stuff' had no electric charge. String theory alters this